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The Ubiquity of Business Groups around the World: A Property-Rights Theory

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ABSTRACT Business groups (BGs) are collections of legally independent firms connected by ownership ties and are widely present in both developing and developed economies. We develop property-rights theory (PRT) to explain this prevalence, extending its scope from the allocation of rights over a bundle of assets to the legal partitioning of assets as another allocative mechanism of property rights. Our conceptual framework complements traditional theories understanding BGs primarily as responses to institutional voids. Drawing on a large global dataset of firms, we examine how firm asset specificity and ownership sharing influence the controlling owners' decision to organize their economic activities as a BG or as a standalone firm. We find that the likelihood of adopting a BG organizational form over a standalone entity increases when (1) controlling owners rely on more firm-specific assets, and (2) ownership is shared or potentially shared. Moreover, our findings are robust across three commonly observed firm strategic configurations: focused, vertically integrated, and diversified. Our study expands the PRT by developing a theory of property-rights architecture, moving forward from the institutional voids perspective to explain BGs as efficient governance responses to asset and ownership complexities, hence contributing also to the strategic management and international business literatures.

Keywords: asset specificity, business groups, organizational form, ownership control, property-rights theory

INTRODUCTION

In this study, we ask why firms choose to organize as a single, standalone legal entity or as a collection of legally independent entities under common ownership, that

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is a business group (BG) (Belenzon et al., 2023).[1] We tackle this question through property-rights theory (PRT), which centres on the rights owners hold over assets, to use, appropriate, and transfer them (Alchian, 1965). It is, however, important to clarify that, as discussed by Foss and Foss (2022, Ch. 4), there exist two strands of PRT. They refer to the first one as old PRT, which encompasses property rights ideas ranging from Coase (1960) through Alchian (1965), Demsetz (1967), Cheung (1969), and Barzel (1997). In this old strand of PRT, it is already acknowledged that rights to a given asset can be held by more than one party, including corporate persons, and allocated across them. The second strand, which Foss and Foss (2022) label new PRT, draws on insights developed by Grossman and Hart (1986) and Hart and Moore (1990). The main question of the new PRT strand is: given a specific asset, who should own it? Thus, the new PRT settles the design decision, ownership, but leaves unanswered a second, analytically distinct question: whether the owned assets are held within one legal entity or partitioned across several, a decision that may in practice be made jointly with the ownership decision. We address this second decision, asking why owners choose to hold a bundle of assets within a single legal entity, a standalone firm, or partition these assets across multiple, legally independent, commonly owned entities, a BG. In sum, we examine how legal partitioning is in itself an allocative mechanism of property rights.

We argue that the exercise of rights is contingent on the organizational form which determines its ownership modularity. In BGs, owners can partition an asset bundle into separately owned entities and assign rights over each entity independently. This is an option that a standalone firm, with one set of rights covering the whole bundle, does not have. However, organizational modularity can be costly and is not valuable on its own; its benefits outweigh its costs only under specific conditions. We identify these conditions by proposing that modularity pays off when: (1) assets are firm-specific and therefore hard to govern by contract, since partitioning lets owners protect and deploy these assets without exposing the rest of the bundle to the same constraints; (2) ownership is, or might be, shared, since partitioning lets owners absorb that sharing at one entity without diluting control over the rest. Hence, we argue that the advantage of ownership modularity is the underlying mechanism explaining the choice of organizational form. Ownership alone is therefore not a sufficient description of how property rights over an asset bundle are organized. Holding ultimate ownership and productive scope constant, legal partitioning itself creates or forecloses specific possibilities for exercising these rights, for example by introducing minority owners selectively into some affiliates but not others, isolating particular claims, transferring discrete bundles of assets, limiting liability at the affiliate level, or reallocating control across entities. Legal partitioning is thus not simply a legal shell wrapped around a pre-existing allocation of rights; it is itself a mechanism, or technology, for structuring those rights.

Importantly, our PRT approach is novel and distinctive from existing theoretical approaches with family resemblances: Transaction cost economics (TCE), agency theory, and institutional Voids. TCE explains the boundaries, or scope, of the firm, that is, whether activities are brought inside or outside the organization through vertical integration, diversification, or focus (Williamson, 1985). Our question is different. Holding the firm's scope as given, we examine what determines the property-rights

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architecture through which that scope is organized. In practice, the two decisions are indeed jointly determined, but we conceptually isolate the ownership-form decision by holding the scope decision fixed, which lets us study it on its own terms. TCE predicts organizational integration whenever transaction risk rises, regardless of how the resulting bundle of assets is owned. Agency theory treats ownership sharing as a source of owners' conflict to be contained through monitoring and incentive-alignment mechanisms. PRT explains when there might be advantages of organizational modularity for owners.

Finally, the institutional voids perspective is the dominant explanation for the presence of BGs. This literature argues that weak market institutions raise the transaction risk firms face, making firms in weak institutional environments more likely to integrate or diversify (Carney and Gedajlovic, 2002; Khanna and Palepu, 2000; Khanna and Yafeh, 2005, 2007; Yiu et al., 2005). However, institutional voids is silent on why BGs are just as common in institutionally advanced economies (Aguilera et al., 2023, 2025) and ubiquitous within and across countries (Colpan et al., 2010). Our PRT-based account explains this ownership-form choice regardless of the institutional context.

We test our novel insight to PRT by comparing standalone firms and BGs while holding the firm's scope constant, across focused, vertically integrated, and diversified strategies alike, which lets us isolate the determinants of firms' organizational form from those determining firm boundaries. Our large global sample spans firms of all sizes, including small as well as organizationally complex BGs, in both developing and developed countries. We restrict our analysis to domestic BGs, that is, those without foreign affiliates, to keep the test of PRT clean, free of the additional considerations that cross-border affiliate structures introduce (i.e., legal requirements forcing the creation of different legal entities to operate in a given country).

Our study makes three important contributions. First, we advance PRT conceptually by developing a theory of property-rights architecture: alongside the design decision PRT already answers, who should own a given asset, we show that a second decision, whether the owned assets are held within one legal entity or partitioned across several, is itself governed by property-rights logic. This is a question TCE does not generate on its own terms, since its transaction-risk logic predicts integration independently of how ownership is subsequently partitioned. Second, our empirical design draws on a large global dataset that overcomes prior data limitations: we include firms in developed and developing countries, of all sizes, private and listed, and we track complete ownership chains, distinguishing full from partial ownership. Third, we contribute to the strategic management and international business literature by moving beyond the view that BGs emerge mainly as a vertically integrated response to institutional voids, and instead we theorize and test the choice of organizational form (BG or standalone), across focused, vertically integrated, and diversified firms alike.

BUSINESS GROUPS AND PROPERTY-RIGHTS THEORY

Alchian (1965) argues that differences in owners' abilities, knowledge, and attitudes towards risk shape the wealth created through the exercise of property rights. Foss

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et al. (2021) similarly show that ownership can be exercised with varying degrees of competence, with direct implications for value creation. According to PRT, owners exert control over the resources they possess through three distinct rights: use, appropriation, and transfer (Foss and Foss, 2005; Kim and Mahoney, 2002). These rights protect owners by limiting others' ability to impose decisions that diverge from their preferences over the assets (Chrisman et al., 2025).

TCE offers a well-established account of firm boundaries: it predicts that activities are internalized or market driven contingent on the hazard of governing a transaction under relationship-specific investments (Foss and Foss, 2022, p. 76; Williamson, 1985). Our argument goes a step further. We take that firm boundary decision, (i.e., focused, vertically integrated, or diversified), as settled, and ask: *what determines the ownership architecture through which the resulting bundle of assets is then held, within one legal entity or partitioned across several*. Our contribution to PRT is, therefore, a theory of property-rights architecture: incorporating organizational form itself, the choice between a standalone entity and a BG, as a second, endogenous property-rights design decision. Thus, TCE operates as a precondition for our contribution rather than as a rival explanation of it.

The BG organizational form offers advantages of flexibility over standalone firms in exercising control over assets through ownership rights. However, these advantages involve higher organizational costs, such as formal governance structures and separate formal information systems for each firm legal entity. Consequently, owners must weigh these costs against the benefits when selecting an organizational form. Drawing on insights from PRT, we argue that two key factors help explain owners' adoption of BGs over standalone firms when governing their ownership rights: (1) the nature of the firm's assets (Hypothesis 1), and (2) whether controlling owners share, or might share, ownership with minority owners (Hypothesis 2). In both cases, partitioning control across multiple entities facilitates the exercise and protects owners' rights to use, appropriate, and transfer assets in ways that concentrating those rights in a single entity does not.

First, the specificity of a firm's assets shapes the effective exercise of ownership rights. Firm-specific assets are valuable basically in their current use and are, therefore, non-easily contractible (Kermani and Ma, 2023). The more easily contractible the assets, the easier it is for owners to determine their use, capture their returns, and transfer them through formal arrangements in market transactions. When assets are highly firm-specific, such as specialized employee training, customized R&D, or a supplier-specific infrastructure, they are more difficult to use, transfer, and appropriate via standard market transactions. Furthermore, in general, firm-specific assets are particularly salient because they underpin competitive advantage (Barney, 1991; Chrisman et al., 2025). In such cases, ownership becomes a critical governance mechanism, and BGs' ownership structures (formed with multiple legally independent firm entities) allow further divisibility and better protection of the property rights over these crucial assets.

Second, the extent to which owners share ownership over assets depends on factors such as risk aversion, flexibility over their financial requirements, and reluctance to dilute control rights. In contexts where owners share, or might share, ownership to maintain strategic flexibility or limit exposure to operational risk, organizing as a BG

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offers strategic advantages. BGs allow for differentiated ownership structures across multiple firm entities while still preserving the core control of property rights by the controlling BG owner at the apex.

In the next subsections, we discuss how firm-specific assets and ownership sharing confer advantages to the BG organizational form, relative to the standalone firm, in the exercise of the property rights to use, appropriate, and transfer firm assets. This organizational form might potentially offset the higher governance and information costs associated with BGs.

Table I summarizes these arguments in two columns and three rows. Column 1 sets out why the BG organizational form offers advantages over the standalone form for exercising each of the three ownership rights, use, appropriation, and transfer, when firm-specific assets are relevant to the economic activity being developed. Column 2 does the same when the controlling owner shares, or wants to share, ownership with other shareholders. Table I is thus a compact map from each of our two theoretical drivers to the specific BG advantage on each right, the logic underlying Hypotheses 1 and 2 below.

Property Rights and Firm-Specific Assets

According to Argyres and Zenger (2012), the diversity of firms' organizational structures arises from ownership relationships, which are in turn shaped by the attributes of the firm's assets and how they are combined within the organization. Foss and Foss (2005, 2022) further contend that joint ownership of non-human, alienable assets can be effectively governed through formal, legally binding contracts, provided that these assets are explicitly contractible. These assets typically have clear market value and are not strongly dependent on specific relational or organizational contexts; examples include office equipment, real estate, vehicles, or generic software licences. Owners of such assets can easily protect their property rights through a single legal entity, a standalone firm.

PRT emphasizes the bundle of rights associated with specific asset characteristics (Foss and Foss, 2005), and stresses that more complex organizational structures become relevant when non-contractible firm-specific investments, such as human capital or relationship-specific investments, complement explicitly contractible assets (Foss and Foss, 2022). From this perspective, the firm consists of both contractible assets and assets requiring specific, non-contractible investments, with ownership arrangements playing a central role in governing their joint use. Owners holding control rights can deploy firm assets in ways that best preserve or enhance their value. PRT emphasizes that such control is critical for mitigating potential losses associated with firm-specific investments, particularly those arising from non-contractibility and relationship-dependence. The degree of asset ownership, whether full or shared, shapes both the investment incentives and the ability to appropriate and transfer surplus. As Hart and Moore (1990) argue, the allocation of property rights depends primarily on the nature of the assets rather than on broader institutional weaknesses.

Thus, PRT suggests that the organizational structure must be aligned to protect and incentivize investments in such firm-specific assets. These assets are particularly important

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Table I. Property rights and key business group factors

BGs & PRT	*Specific assets*	*Ownership control*
Right to use	• BGs distribute firm-specific assets among different affiliates, isolating the controlling owner's right to use relevant resources from minority shareholders and stakeholders of other affiliates. • Firm-specific assets, such as specialized knowledge or human capital developed in one affiliate, may be redeployed through ownership-based coordination mechanisms in another affiliate with its specific minority shareholders and stakeholders.	• BGs enhance financing possibilities, facilitating new equity issuance while preserving control for the controlling shareholder and increasing debt capacity through reduced default risk. • In a BG, the controlling owner controls asset use across affiliates, enabling strategic resource allocation and coordination, and maintaining operational flexibility with limited capital investment.
Right to appropriate	• Firm-specific assets can be allocated across affiliates with different minority owners and stakeholders, fine-tuning the distribution of rents. • The internal capital markets of BGs generate flexibility in the distribution of rents from relevant assets (i.e., firm-specific) across different affiliates.	• BGs can operate under ownership structures that separate control from cash-flow rights (i.e., pyramidal BGs). • BG design allows leveraging the limited liability of firms and the asymmetric appropriation of profits and losses.
Right to transfer	• Firm specificity limits asset redeployability and reduces liquidity; BGs mitigate this by enabling the transfer of firm-specific assets through the sale of affiliated firms. • Exit or switching real options are more valuable in BGs since specific assets may be sold jointly with the remaining production inputs attached to a given affiliate.	• BGs offer greater strategic flexibility in transferring assets, due to the modularity of their legal structure. Affiliated firms can be sold, restructured, or partially floated without affecting the entire group. • The BG structure allows partial asset transfers that preserve voting power or influence in the remaining parts.

because they are central to the creation of competitive advantage (Barney, 1991). Therefore, controlling owners have incentives to adopt the organizational form (BG vs standalone firm) that better protects their property rights over these assets vis-à-vis minority owners and other stakeholders. Accordingly, we compare the BG and the standalone firm organizational choices in terms of how each enables the exercise of the three property rights over firm-specific assets, as summarized in column 1 of Table I:

Rights to use. In standalone firms, centralized control enables owners to align the deployment of their firm-specific assets with their long-term strategic objectives. However, when minority owners, or even other stakeholders, can influence use rights, the standalone form may constrain the controlling owner's flexibility to reallocate

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such assets across business units in rapidly changing environments (Gibbons, 2005). By contrast, BGs can distribute use rights across legally distinct entities, enabling the modular deployment of firm-specific assets to affiliates while insulating them from the claims of minority owners and stakeholders at the BG level. This modularity concentrates control in the hands of the BG's controlling owner and enhances strategic flexibility, helping explain why some entrepreneurs rely on BGs to grow and diversify without relinquishing control (Iacobucci and Rosa, 2010). Moreover, firm-specific assets, such as specialized knowledge or human capital developed in one business affiliate within the BG, can be redeployed to another through ownership-based coordination mechanisms, allowing BGs to adapt asset use across affiliates while preserving control (Argyres et al., 2012; Foss and Foss, 2022).

Rights to appropriate. The returns from firm-specific investments in standalone firms accrue directly to the firm and its owners, ensuring strong internal incentive alignment. In contrast, within BGs, firm-specific assets may be allocated across different legal entities with different minority owners and stakeholders. The BG organizational form allows the controlling owner to determine which entities, and therefore which sets of minority owners and stakeholders, hold appropriation rights over particular assets. Furthermore, appropriation rights in BGs can be redistributed across affiliated firms via internal capital markets, profit sharing, or transfer pricing (Almeida and Wolfenzon, 2006; Khanna and Yafeh, 2007). This organizational flexibility supports cross-subsidization and strategic coordination across multiple entities.

Rights to transfer. The transfer of firm-specific assets is difficult because their value is context-dependent and often not easily separable or marketable (Gibbons, 2005; Kermani and Ma, 2023). These traits limit firms' redeployment options and reduce asset liquidity. BGs facilitate the transfer of non-easy contractible assets by reallocating ownership rights across firms. By shifting affiliate ownership, BGs enable the joint reallocation of specific assets that cannot be efficiently governed through contracts, together with complementary production factors. This ability can be interpreted as switching real options to exit or redeploy resources (Chi et al., 2019), which entails access to resources via control over affiliated firms, giving BGs an organizational advantage in reallocating complex, firm-specific assets (Foss and Foss, 2022; Langlois, 2003).

Summarizing, as also synthesized in Table I, although the BGs organizational choice has higher governance and information costs than the standalone firm, our discussion shows that the BG organizational form grants key advantages to the controlling owner to exercise and protect its property rights over firm-specific assets. Therefore, when the proportion of firm-specific assets (relevant assets for the strategy of value creation, Williamson, 2025) is high, we expect that the benefits of the BG organizational choice will overcome its costs, as formally stated in the following hypothesis:

> *Hypothesis 1:* When the proportion of firm-specific assets is high, the business group organizational form is more likely to be adopted over a standalone organizational form to facilitate the exercise of property rights.

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Property Rights and Ownership Sharing

Ownership-related considerations regarding control, risk exposure, and the degree of shared ownership influence whether they are organized as standalone entities or with BGs. BGs, particularly in developing economies, feature governance mechanisms, such as pyramidal ownership, crossholdings, and partial or full ownership arrangements, which enable a separation between control and cash-flow rights (Almeida and Wolfenzon, 2006). These mechanisms allow controlling owners to retain effective control while developing their strategies, diversifying risk, or leveraging group-wide resources while minimizing equity. In contrast, standalone firms may lack the explicit internal capital and monitoring devices of BGs, and this often requires controlling parties to make higher investments to safeguard resource utilization and ensure exclusive appropriation of residual rents. Indeed, when controlling owners share, or might share, ownership, the BGs organizational form provides the owners with several advantages over the standalone firm for exercising and protecting their property rights. A summary of the arguments can be followed in column 2 of Table I for every property right, as we discuss below:

Rights to use. Owners of standalone firms who retain full control over their assets' use achieve strong alignment between ownership and control, but they also have to bear concentrated risk and managerial responsibility. According to Hart and Moore (1990), when ownership is shared or partially transferred, the control rights over firm assets become diluted, potentially weakening the original owner's ability to direct strategic decisions, in particular, when contracts are incomplete and external investors are introduced. In BGs, rights to use are often centralized at the top of the BG organizational form, while affiliated firms may face reduced autonomy, as strategic decisions are subordinated to group-level coordination, which can facilitate resource pooling (Guzzini and Iacobucci, 2014). In a BG, the parent firm determines the assets' use across affiliated firms even without full ownership. BGs carry out strategic coordination across affiliated firms while maintaining decision-making flexibility of the controlling owner. Moreover, the limited-liability status of each affiliate implies that when assets are allocated across multiple industries within a BG, the group effectively diversifies its exposure to industry-specific shocks. This cross-affiliate diversification mitigates aggregate default risk for the BG as a whole (Heaney and Holmen, 2008). This default risk mitigation might endow BGs with higher debt capacities than otherwise similar standalone peers, representing a substantial advantage as documented by Bianco and Nicodano (2006). Through pyramidal structures, owners can control multiple affiliated firms with a small investment, preserving their rights to use while reducing exposure to affiliated firm-specific risk (Almeida and Wolfenzon, 2006).

Rights to appropriate. There is greater alignment between effort and reward when owners in standalone firms hold both control and cash-flow rights. This reduces the risk of value diversion and protects minority owners (if any) from opportunistic behaviour. The appropriation of returns in a standalone firm is typically straightforward, that is, profits

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accrue directly to the owners. However, when controlling owners share ownership, they face a trade-off between financing growth and diluting their claims on future profits, something typically seen in family-run firms.

This is when BGs organizational form offers advantages to exercise the appropriation rights. BGs can operate under ownership structures that separate control from cash-flow rights, sometimes in a pyramidal structure (Almeida and Wolfenzon, 2006). Controlling owners can retain control without holding full ownership, creating opportunities for tunnelling and related-party transactions (Morck et al., 2005). Additionally, BGs can leverage the limited liability of affiliated firms (Belenzon et al., 2023), enabling asymmetric appropriation of profits and losses (i.e., not compensating losses from specific affiliates with profits from the remaining affiliates in the BG). Almeida and Wolfenzon (2006) show that ownership structure is related to liability exposure, allocating ownership rights to limit the extent of risk borne by the owner. In incomplete-contract settings, risk coverage creates a need for ownership to secure control and protect the owner's specific investments.

Rights to transfer. Transferring assets or ownership in standalone firms often entails selling the entire firm or divesting specific business units. Such moves may introduce significant market frictions, a loss of synergies, or exposure to market scrutiny. Conversely, BGs offer greater strategic flexibility in transferring assets due to the modular nature of their legal structure. Affiliated firms can be sold, restructured, or partially floated without affecting the entire BG (Kermani and Ma, 2023). This enables owners to manage portfolio risk, respond to market conditions, or raise capital while maintaining control. Moreover, the BG organizational form allows partial asset transfers that preserve voting power or influence, especially in pyramidal hierarchies (Morck et al., 2005), which spread risk across affiliated firms.

Overall, this discussion, also synthesized in Table I, shows how the BG organizational choice offers greater flexibility for exercising and protecting the property rights of the controlling owners, which is especially relevant when they share, or might share, ownership. Therefore, our second formal hypothesis is as follows:

> *Hypothesis 2:* Controlling owners sharing, or who might share, ownership are more likely to adopt a business group over a standalone organizational form to protect their property rights.

METHODS AND DATA

Although most of the studies of BGs focus on vertical integration strategies (e.g., Atalay et al., 2014), some examine horizontally diversified BGs (e.g., Khanna and Yafeh, 2007), while others document cases in which BGs narrow the operational scope to a single industry (Khanna and Rivkin, 2001). We explore the full spectrum of strategic configurations: focused, vertically integrated, and diversified, when explaining the observed presence of BGs across different institutional contexts. This is consistent with our theoretical approach seeking to test our argument on the organizational form decision once the boundaries of the firm – strategic configurations – are

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determined. Thus, our empirical design classifies both standalone firms and BGs into these three strategic configurations, allowing us to isolate organizational form (BG vs. standalone) from strategic configuration.

We use data from Moody's ORBIS database that contains information about public and private firms all around the world. In the first step of our analysis, we identify BGs based on ownership ties. There are other broader operationalizations of BGs based on informal ties (Guillén, 2000; Yiu et al., 2005). However, given our interest in property rights, we focus on ownership ties. Thus, we follow previous literature that identifies BGs based on ownership (e.g., Aguilera et al., 2020; Belenzon et al., 2023; Faccio et al., 2021; Faccio and O'Brien, 2021; Masulis et al., 2011).

Sample Selection

For our BG identification process, we consider all firms with data on ownership in the ORBIS database, that is, 9,662,352 firms from 204 countries in 2020. In line with the criteria established in previous studies, we define control as the largest shareholder owning more than 50 per cent of the shares in private firms or more than 20 per cent in publicly traded companies (Belenzon et al., 2019). Using this approach, we identified 1,464,801 firms affiliated with 373,097 BGs.

Our final sample includes all standalone firms and all firms affiliated with BGs, with data aggregated at the BG level. Specifically, we consolidate information for BGs, from both parent and affiliated firms (i.e., we measure total assets as the sum of assets across all entities within the group). As a result, each BG is represented by a single observation. This approach ensures that the unit of analysis is the individual firm for standalone entities and the entire BG for affiliated firms, enabling direct and meaningful comparisons between the two. To maintain consistency when comparing standalone firms and BGs, we restrict the sample to entities that operate exclusively within a single country, deliberately excluding BGs with foreign affiliates. This eliminates cases where BGs may have been created primarily for regulatory reasons related to cross-border investments.

We further refine our sample by retaining only firms with non-missing data on key variables required for our analysis: our measures of firm-specific assets, total assets, shareholder ownership, country, and NACE industry classification codes. We also exclude both standalone firms and BGs with fewer than ten employees, consistent with the definition of micro-enterprises established by the Commission Delegated Directive (EU) 2023/2775 of 17 October 2023. The resulting dataset comprises 684,547 organizations (both BGs and standalone firms) from 88 countries and 88 industries (classified at the 2-digit NACE level), of which 30,355 are BGs.

Using the 2-digit NACE codes, we classify both standalone firms and BGs according to strategic type, that is, whether they operate in a single segment (focused) or multiple segments. Among multi-segment observations, we distinguish between vertically integrated and diversified firms, that is, those which are not vertically integrated.

For standalone firms, we use all reported 2-digit NACE codes (primary and secondary). We calculate vertical integration intensity as the proportion of industry pairs within the firm that are typically involved in supplier-customer relationships relative to

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the total number of possible industry combinations. For BGs, we apply the same logic but use each affiliate's primary 2-digit NACE code. We first enumerate all possible industry code pairs within the group and then calculate the share of these pairs that represent supplier-customer links. To determine typical supplier-customer relationships between industries, we use the 2020 US Input–Output Tables in concordance with OECD industry codes. For each industry, we identify the five largest customer industries as those with the highest input purchases from that industry. We map our affiliates' 2-digit NACE codes to the OECD input–output industry classification using the OECD concordance, and then merge the firm- and group-level industry data with the input–output tables. We then construct our vertical integration measure at the business-group level by counting the number of affiliates that operate in industries classified as upstream or downstream of the group's main industry (based on these input–output linkages), normalized by the total number of potential links within the group. This measure captures vertical integration as common ownership of upstream and/or downstream affiliates along the input–output production structure, although it does not directly measure within-group intermediate-goods flows.[2] Our approach, using industry-level production linkages to infer firm- or group-level vertical integration, follows the spirit of Kermani and Ma (2023), who build on the input–output framework of Acemoglu et al. (2009) and construct upstreamness/downstreamness scores by weighting subsidiaries with input–output coefficients.

Based on this methodology, we classify all the BGs and standalone firms into three strategic types: (1) *single segment: focused* (i.e., BGs or standalone firms) which operate only in one industry; (2) *multiple segments: vertical integration* which includes organizations operating in multiple industries with at least one supplier-customer tie among them, indicating vertical integration; and (3) *multiple segments: diversification*, which includes organizations in multiple industries but no vertical ties.

Table II presents the distribution of standalone firms and BGs by strategic type. The *Small*, *Medium*, and *Large* categories refer to organizations with a total number of employees between 10–49, 50–249, and 250 or more, respectively, as defined by the Commission Delegated Directive (EU) 2023/2775. We group countries according to the World Bank's classification: 'High-Income' countries and 'Low-Income' which include upper-middle, lower-middle, and low-income economies. We observe that most observations are focused standalone firms that operate in a single segment, particularly among small firms. BGs are more present in the 'Multiple segments: Vertical Integration' and 'Diversification' categories, especially among medium and large firms. High-income countries account for the majority of observations, while low-income countries have fewer because of the lower coverage of firms from these regions in the ORBIS database. Vertical integration appears more prevalent among larger BGs. These figures suggest that both size and national economic context influence the choice between operating as a BG or as a standalone firm.

Variable Definitions and Descriptive Statistics

Firm-specific assets. To explore the determinants influencing the organizational form adopted – BG versus standalone – a central explanatory variable as discussed in H1 is firm-specific assets, which we operationalize following Kermani and Ma (2023). Kermani and Ma (2023)

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Table II. Firms by organizational form and strategy type

Strategy type	*Single segment: Focused*		*Multiple segments: Vertical integration*		*Multiple segments: Diversification*	
Organizational form	*BG*	*Standalone*	*BG*	*Standalone*	*BG*	*Standalone*
Total	10,145	364,803	10,817	180,957	9393	108,432
By size						
Small	4819	289,368	4285	125,158	4408	79,311
Medium	3827	66,579	3413	48,935	3120	26,059
Large	1499	8856	3119	6864	1865	3062
By country						
High-income	8640	315,918	9321	66,758	7632	57,302
Low-income	1505	48,885	1496	114,199	1761	51,130

Note: This table reports the sample distribution across three organizational-strategic categories. Organizations are classified as BGs or standalones upon their strategic configuration: Single-segment, multi-segment with vertical integration, and multi-segment with diversification. The sample is further disaggregated by firm size (small: 10–49 employees; medium: 50–249; large: 250+) and country income level using World Bank definitions. The unit of analysis is BG and standalone firm. All firms included in a BG operate in a single country, and BGs and standalone firms have at least ten employees.

focus on the proportion of the accounting value of property, plant, and equipment (PPE) that may be recovered in the case of liquidation (within 6 to 12 months) for different industries in the United States. More specifically, we measure firm-specific assets with the variable RECOVERY_PPE, defined as (PPE/Total Assets) × PPE_RecoveryRate, where PPE_RecoveryRate is obtained from the industry-level PPE recovery rates reported by Kermani and Ma (2023). For each firm, we assign the recovery rate of every 2-digit NACE code it reports (primary and secondary). For standalone firms, PPE_RecoveryRate is the average of these rates across the firm's reported codes. For BGs, it is the average across the codes reported by the parent plus its affiliates, so that the same aggregation rule is applied to the relevant decision unit in both cases.[3]

We acknowledge, however, that there exist alternative proxies of firm-specific non-easily contractible assets. To address this concern, we include robustness tests using two alternative proxies. Specifically, we consider *Intangible Assets* (INTANGIBLES) (e.g., customer data and goodwill), which are typically harder to redeploy than tangible assets. Additionally, following the logic proposed by Kermani and Ma (2023), we also include a robustness test using the *Speed of Depreciation* (SPEED DEPRECIATION), as an alternative proxy of firm-specific assets, calculated as the ratio of annual depreciation and amortization expenses to fixed assets. Faster-depreciating assets are assumed to be less redeployable and thus more firm-specific. These complementary measures help reinforce the interpretation of our results and reduce concerns regarding construct validity in our test of asset specificity.

Ownership. We operationalize controlling owner's share of ownership by the equity stake held by the controlling owner. For standalone firms, this refers to the ownership share in the firm itself, while for BGs, it refers to shareholding in the parent company. To capture this dimension, we construct two variables: ID(%Owner = 100%), a binary

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indicator equal to one when the controlling owner holds 100 per cent of the equity, and % OWNERSHIP, the continuous percentage of equity held.

We control for firm size by using the natural logarithm of total assets (ln*ASSETS*) and the number of employees (*EMPLOYEES*). We account for industry heterogeneity through fixed effects based on the firm's primary 2-digit NACE industry code.

Table III presents descriptive statistics for the variables used. Consistent with our data in Table II, standalone firms are the vast majority of observations, while BGs represent only 2.7 per cent, 5.2 per cent, and 8.0 per cent of firms in the Focused, Vertical Integration, and Diversification strategy categories, respectively.

The proxy for firm-specific assets, *RECOVERY PPE*, is similar across all strategy categories, with slightly lower values in the diversified category (0.084). The intangibility ratio of organizations with focused strategy shows a slightly higher mean (0.009), possibly due to a higher concentration in industries where proprietary knowledge matters more. Organizations with vertically integrated and diversified strategies have similar mean intangibility levels (0.004 and 0.005), though dispersion is modest in all groups. As for Speed of Depreciation, we observe that focused organizations exhibit the highest average depreciation speed (0.196), implying a five-year amortization horizon. Diversified organizations follow with a mean of 0.095, while vertically integrated ones show the slowest rate (0.08), indicating relatively more durable and potentially less specific and more contractible capital assets.

The ownership structure varies by organization's strategy. Full ownership by the controlling owner is particularly common in vertically integrated and diversified organizations (71 per cent), relative to 58 per cent in focused organizations. Mean ownership levels are high across all strategy categories, exceeding 87 per cent.

In terms of the number of employees, vertically integrated organizations are considerably larger, with a mean of 111 employees, nearly double the 58 observed in the focused strategy category. Diversified organizations fall in between, averaging 80 employees. As for our other measure of organizational size, lnASSETS, we observe differences that are not large in magnitude, suggesting that organizations with more complex structures are not necessarily larger in terms of their asset base.

Table IV presents the correlation matrix between our dependent variable, BG (dummy variable identifying when an organization takes the BG structure), and the explanatory variables used in our analysis. The correlation between BG and Recovery PPE (asset specificity) is negative and statistically significant, in line with our first hypothesis. The lower recoverable PPE value in the case of liquidation (suggesting the presence of firm-specific non-easily contractible assets), the BG organizational form is more likely. Intangibles and Speed of Depreciation show a positive and significant correlation with BG, as expected, as proxies for investment specificity. Notably, Intangibles and Speed of Depreciation are themselves positively and significantly correlated, and negatively correlated to Recovery PPE, reinforcing the interpretation that these variables capture dimensions of asset specificity. We observe a positive correlation between BG and size (LnASSETS), suggesting that BGs tend to be larger than standalone firms. To address this size effect in our empirical analysis, we stratify the sample into three groups of comparable size based on the number of employees (EMPLOYEES). BG is negatively correlated with the two ownership variables, which is consistent with Hypothesis 2.

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Table III. Descriptive statistics by strategy types

	Average	*Std.Dev.*	*Median*	*P5th*	*P25th*	*P75th*	*P95th*
Single segment: Focused N = 374,948							
BG	0.027	0.162	0.000	0.000	0.000	0.000	0.000
EMPLOYEES	58	494	25	11	16	44	161
RECOVERY PPE	0.088	0.095	0.056	0.000	0.015	0.130	0.284
lnASSETS	7.757	1.554	7.728	5.318	6.834	8.659	10.339
% OWNERSHIP	0.873	0.197	1.000	0.510	0.750	1.000	1.000
ID (%Owner = 100%)	0.578	0.494	1.000	0.000	0.000	1.000	1.000
INTANGIBLES	0.009	0.023	0.000	0.000	0.000	0.004	0.090
SPEED DEPRECIATION	0.196	0.245	0.111	0.000	0.002	0.277	0.826
Multiple segments: Vertical integration, N = 191,774							
BG	0.052	0.221	0.000	0.000	0.000	0.000	1.000
EMPLOYEES	111	2053	34	13	21	61	253
RECOVERY PPE	0.087	0.098	0.049	0.000	0.007	0.139	0.291
lnASSETS	7.282	2.038	7.221	4.070	6.074	8.368	10.681
% OWNERSHIP	0.906	0.177	1.000	0.510	0.900	1.000	1.000
ID (%Owner = 100%)	0.713	0.452	1.000	0.000	0.000	1.000	1.000
INTANGIBLES	0.004	0.015	0.000	0.000	0.000	0.000	0.024
SPEED DEPRECIATION	0.080	0.181	0.000	0.000	0.000	0.076	0.441
Multiple segments: Diversification, N = 117,825							
BG	0.080	0.271	0.000	0.000	0.000	0.000	1.000
EMPLOYEES	80	497	31	12	19	56	217
RECOVERY PPE	0.084	0.094	0.050	0.000	0.009	0.132	0.278
lnASSETS	7.568	1.832	7.548	4.581	6.480	8.613	10.640
% OWNERSHIP	0.910	0.173	1.000	0.510	0.909	1.000	1.000
ID (%Owner = 100%)	0.712	0.453	1.000	0.000	0.000	1.000	1.000
INTANGIBLES	0.005	0.016	0.000	0.000	0.000	0.001	0.030
SPEED DEPRECIATION	0.095	0.194	0.000	0.000	0.000	0.109	0.495

Note: This table reports descriptive statistics for the three strategic groups defined in our sample: Variables include: BG (dummy equal to 1 for firms belonging to a business group), number of employees, *RECOVERY PPE* (ratio of PPE over total assets multiplied by the recovery rate of the accounting value PPE investments in the case of liquidation as in Kermani and Ma, 2023), lnASSETS (log of total assets), % *OWNERSHIP* (share of ownership held by the largest shareholder), *ID(%Owner = 100%)* (indicator equal to 1 if the largest shareholder owns 100% of shares), *INTANGIBLES* (ratio of intangible to fixed assets), and *SPEED DEPRECIATION* (ratio of depreciation expenses to total fixed assets). Columns report the mean, standard deviation, median, and selected percentiles ($P5^{th}$, $P25^{th}$, $P75^{th}$, and $P95^{th}$).

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Table IV. Correlation matrix

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1)	BG	1.000							
(2)	RECOVERY PPE	−0.023***	1.000						
(3)	% OWNERSHIP	−0.097***	−0.045***	1.000					
(4)	ID(%Owner = 100%)	−0.072***	−0.052***	0.828***	1.000				
(5)	lnASSETS	0.321***	0.139***	−0.156***	−0.150***	1.000			
(6)	EMPLOYEES	0.071***	0.011***	−0.011***	−0.003*	0.103***	1.000		
(7)	INTANGIBLES	0.064***	−0.062***	−0.079***	−0.090***	0.116***	0.011***	1.000	
(8)	SPEED DEPRECIATION	0.057***	−0.262***	−0.079***	−0.089***	0.006***	−0.007***	0.263***	1.000

Note: This table reports Pearson pairwise correlations among the key variables used in our analysis. BG is a dummy equal to 1 if the firms belong to a business group. RECOVERY PPE is the ratio of PPE over total assets multiplied by the recovery rate of the accounting value PPE investments in the case of liquidation as in Kermani and Ma (2023); ln *ASSETS* is the natural logarithm of total assets; % *OWNERSHIP* is the share of equity held by the largest shareholder; *ID(%Owner = 100%)* is a dummy equal to 1 if the largest shareholder owns 100% of the firm; *INTANGIBLES* is the ratio of intangible to total assets; and *SPEED DEPRECIATION* is the ratio of depreciation to total fixed assets. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Empirical Strategy

To test Hypothesis 1, we estimate the following baseline model:

$$BG_i = \gamma_0 + \gamma_1 RECOVERY\,PPE_i + \gamma_2 \ln Assets_i + \eta_i^S + \eta_i^c + \epsilon_i \quad (1)$$

where η_i^S denotes industry fixed effects, captured using each organization's primary 2-digit NACE code as reported in the ORBIS database, and η_i^c represents country-fixed effects, based on the location of the standalone firm or the parent company in the case of BGs. These fixed effects help absorb unobserved heterogeneity associated with systematic differences in organizational forms or asset composition across industries and countries. In particular, industry effects might control for their typical technological intensity or capital structure, while country effects might account for institutional, regulatory, or macroeconomic factors that may influence firm structure and governance. ϵ_i is the error term. The dependent variable, BG, is measured as of 2020, and the independent and control variables, *RECOVERY PPE* and ln*ASSETS*, are lagged to 2017 values to (at least partially) mitigate endogeneity concerns arising from the potential simultaneity between the organization's characteristics and the organizational form (i.e. BG and standalone firm). We clustered standard errors at the country level to account for within-country correlation.

Our results will support Hypothesis 1 if γ_1 is negative and statistically significant, that is, the more firm-specific investments captured by a low recovery rate of the accounting value of PPE in case of liquidation, the higher the probability of observing a BG.

To test Hypothesis 2, we estimate a multinomial logit model, a standard approach for modelling discrete choices among more than two unordered alternatives. In

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our case, the dependent variable, denoted as $OrgForm_i$, captures the organizational form among standalone and BGs, distinguishing fully owned from partially owned BGs. This variable can take one of three values: (1) standalone firm, (2) BG with fully owned affiliates (100% OWNED BG), and (3) BG with partially owned affiliates (PARTIALLY OWNED BG). Although all BGs share a common coordination logic, differences between fully and partially owned structures are capturing how ownership control influences organizational forms. Fully owned BGs feature more centralized control and stronger internal enforcement, whereas partially owned BGs imply greater capital flexibility and access to external investors, albeit with diluted control rights.

The multinomial logit model estimates the relative probability of adopting each organizational form (fully or partially owned BG) compared with standalone firms as the reference category. The model estimates separate coefficients for each explanatory variable across the two BG categories. These coefficients reflect the change in the log-odds of adopting that organizational form (relative to being a standalone firm) associated with a one-unit change in the covariate, holding all else constant.

Accordingly, we estimate the following model:

$$\Pr\left(OrgForm_i = j\right) = \frac{\exp\left(\beta_{j0} + \beta_{j1} RECOVERY\,PPE_i + \beta_{j2} \ln Assets_i + \beta_{j3} Ownership_i + \eta_i^S + \eta_i^c\right)}{\sum_{k=1}^{3} \exp\left(\beta_{k0} + \beta_{k1} RECOVERY\,PPE_i + \beta_{k2} \ln Assets_i + \beta_{k3} Ownership_i + \eta_i^S + \eta_i^c\right)} \tag{2}$$

where j identifies the type of organizational form (1, 2, or 3), and Ownership is a variable that captures the proportion of shares held by the controlling shareholder, *ID(Owner = 100%)*, and *%Ownership*. Our main specification uses *ID(Owner = 100%)*, while we include *%Ownership* in robustness tests as a continuous alternative. η_i^S denotes industry fixed effects, and η_i^c represents country-fixed effects. As in Hypothesis 1, the dependent variable is measured as of 2020, while all independent and control variables are lagged to 2017 (t − 3) values to partially mitigate endogeneity concerns. We cluster standard errors at the country level to account for within-country correlation.

A negative and statistically significant coefficient β_{j1} for Recovery PPE for category $j = 2$ (fully owned BG) implies that higher firm-specific investments increase the relative likelihood of being in a fully owned BG compared with being a standalone firm. Similarly, a negative coefficient for Ownership for category $j = 3$ (partially owned BG) would suggest that firms with more dispersed ownership are more likely to adopt partially owned BG structures relative to remaining standalone.

RESULTS

Table V serves to identify the relationship between firm-level asset specificity and the adoption of a BG versus a standalone firm (H1). The main explanatory variable, *RECOVERY PPE*, proxies for firm-specific assets. Panel A details results for the full sample of countries.

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Table V. Determinants of organizational from and firm-specific assets

	Single segment: Focused			*Multiple segments: Vertical integration*			*Multiple segments: Diversification*		
	Small	*Medium*	*Large*	*Small*	*Medium*	*Large*	*Small*	*Medium*	*Large*
Panel A: Total countries									
RECOVERY PPE$_{t-3}$	−0.069*** (0.011)	−0.172*** (0.029)	−0.265*** (0.093)	−0.112* (0.063)	−0.199* (0.113)	−0.384*** (0.121)	−0.127** (0.053)	−0.192* (0.111)	−0.051 (0.193)
lnASSETS $_{t-3}$	0.012*** (0.002)	0.025*** (0.005)	0.043***(0.007)	0.018* (0.010)	0.027* (0.014)	0.068*** (0.018)	0.027** (0.011)	0.039** (0.016)	0.075*** (0.023)
N. Observations	294,187	70,406	10,355	129,443	52,348	9983	83,719	29,179	4927
R^2	0.095	0.148	0.187	0.386	0.416	0.575	0.361	0.359	0.385
Panel B: High income									
RECOVERY PPE$_{t-3}$	−0.078*** (0.013)	−0.205*** (0.025)	−0.333*** (0.084)	−0.257*** (0.060)	−0.484*** (0.089)	−0.579*** (0.107)	−0.214*** (0.039)	−0.433*** (0.055)	−0.407*** (0.118)
lnASSETS $_{t-3}$	0.013*** (0.003)	0.028*** (0.007)	0.046*** (0.010)	0.051*** (0.011)	0.080*** (0.014)	0.101*** (0.013)	0.056*** (0.007)	0.087*** (0.007)	0.116*** (0.018)
N. Observations	258,472	57,711	8358	53,649	17,745	4673	48,294	13,725	2898
R^2	0.080	0.123	0.177	0.472	0.481	0.525	0.433	0.423	0.402
Panel C: Low income									
RECOVERY PPE$_{t-3}$	−0.016 (0.010)	−0.055 (0.057)	−0.116 (0.119)	−0.019 (0.015)	−0.057* (0.033)	−0.180*** (0.013)	−0.022 (0.023)	−0.006 (0.026)	0.242 (0.219)
lnASSETS $_{t-3}$	0.008*** (0.000)	0.016*** (0.001)	0.031*** (0.002)	0.006*** (0.001)	0.010*** (0.002)	0.041*** (0.002)	0.011*** (0.002)	0.019*** (0.003)	0.038*** (0.011)
N. Observations	35,709	12,690	1982	75,792	34,602	5306	35,420	15,451	2022
R^2	0.240	0.316	0.284	0.076	0.112	0.191	0.131	0.162	0.237

Note: This table reports results from linear regression models estimating the association between firm-specific capital (proxied by RECOVERY PPE at *t*–3) and the probability of BG affiliation across different strategic configurations: Single-segment, multi-segment with vertical integration, and multi-segment with diversification. We estimated regressions separately by firm size (small: 10–49 employees; medium: 50–249; Large: 250+) and by country income level (total, high-income, and low-income samples). All models include fixed effects for the industry (NACE 2-digit) and country, and we cluster standard errors at the country level to account for intra-country correlation. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

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We find strong empirical support for Hypothesis 1. Firms with higher asset specificity (low *RECOVERY PPE*) are significantly more likely to adopt a BG structure. We also observe some evidence suggesting that the effect of *RECOVERY PPE* becomes more pronounced as firm size increases. Across all strategic categories (focused, vertically integrated, and diversified), the probability of adopting a BG structure decreases with recovery PPE, except for large diversified firms. This pattern suggests that larger organizations, which often engage in more complex operations, may face greater challenges when managing property rights over firm-specific assets, challenges that the BG organizational form is well-suited to address.

Panel B focuses on high-income countries and confirms, with stronger evidence, the association between asset specificity and BG organizational form (negative coefficient of *RECOVERY PPE*). The effect is particularly strong among medium and large firms. For instance, among large vertically integrated firms in high-income contexts, the coefficient for *RECOVERY PPE* is −0.579 ($p < 0.01$), highlighting the role of BGs in managing the property rights of specific assets within complex production structures.

Panel C reports the results for lower-income countries, where the relationship is less pronounced. Among large-sized organizations, the coefficients for *RECOVERY PPE* are substantially larger, although statistically significant only in the case of vertical integration (0.180, $p < 0.01$). This suggests that the primary driver of BG formation in these settings is not specialization per se but rather the need to mitigate market failures through strategic diversification and vertical integration, supporting the institutional voids argument. When markets are weak, the contractability of any asset is affected, not only of firm-specific assets.

Taken together, the evidence shown in Table V indicates that asset specificity is a key determinant of BG adoption, with its effect magnified in institutional contexts where external market mechanisms are reliable.

Table VI provides empirical evidence to evaluate Hypothesis 2, which posits that the allocation of ownership rights reflects the need to protect property rights in different organizational forms. Specifically, our hypothesis predicts that controlling owners who share or might share ownership are more likely to choose a BG over a standalone firm. We estimate two models of BG ownership ties. The first is fully owned, referring to 100 per cent ownership over all parent-affiliate ties. Conversely, partially owned affiliates capture BGs where some parent-affiliate ownership ties are under 100 per cent.

To test this, we estimate model (2), a multinomial logit model in which the dependent variable categorizes organizations as standalone firms, fully owned BGs, or partially owned BGs. The reference group is standalone firms, and the key explanatory variable is ID(%Owner = 100%), a binary indicator equal to one when the controlling shareholder owns 100 per cent of the equity in the standalone firm or in the BG parent company.

The results are strongly consistent with Hypothesis 2. Across all strategic categories and firm sizes, the coefficients for *ID(%Owner = 100%)* are negative and statistically significant, with similar magnitudes for partially owned BGs as for fully owned ones.

These results indicate that firms in which the controlling owner holds 100 per cent of equity are significantly less likely to operate as BGs than as standalone firms. This is in line with the argument that when ownership is not shared with minority owners, standalone firms are more prevalent as a means of safeguarding property rights, while BGs'

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Table VI. Determinants of organizational form and ownership control

Reference group: Standalone	*Single segment: Focused*			*Multiple segments: Vertical integration*			*Multiple segments: Diversification*		
	Small	*Medium*	*Large*	*Small*	*Medium*	*Large*	*Small*	*Medium*	*Large*
100%-OWNED BG									
RECOVERY PPE_{t-3}	−5.206*** (0.561)	−4.575*** (0.693)	−2.508*** (0.845)	−5.493*** (1.548)	−5.416*** (1.844)	−4.082*** (0.900)	−4.010*** (1.090)	−3.510** (1.514)	−1.108 (1.795)
$lnASSETS_{t-3}$	0.818*** (0.080)	0.571*** (0.066)	0.418*** (0.067)	1.356*** (0.100)	1.436*** (0.086)	1.342*** (0.108)	1.288*** (0.118)	1.142*** (0.086)	0.876*** (0.058)
ID(%Owner = 100%)	−20.846*** (0.300)	−20.297*** (0.336)	−19.825*** (0.336)	−20.370*** (0.461)	−21.751*** (1.206)	−21.386*** (0.551)	−20.667*** (0.637)	−20.546*** (0.351)	−20.727*** (0.275)
PARTIALLY OWNED BG									
RECOVERY PPE_{t-3}	−4.288*** (0.365)	−3.888*** (0.495)	−2.783*** (0.702)	−2.274 (1.414)	−2.050 (1.667)	−0.839 (0.916)	−2.418*** (0.850)	−2.296** (0.968)	0.771 (1.427)
$lnASSETS_{t-3}$	0.801*** (0.084)	0.459*** (0.072)	0.348*** (0.061)	1.097*** (0.097)	1.058*** (0.092)	0.857*** (0.086)	1.178*** (0.123)	0.898*** (0.107)	0.590*** (0.057)
ID(%Owner = 100%)	−20.820*** (0.300)	−20.611*** (0.348)	−19.695*** (0.379)	−21.093*** (0.536)	−20.954*** (0.472)	−21.260*** (0.266)	−30.094*** (0.578)	−20.793*** (0.398)	−20.848*** (0.302)
N. Observations	227,390	50,908	7137	74,758	33,407	6576	56,627	20,198	3384

Note: This table presents results from multinomial logit models estimating the likelihood of firms adopting different organizational forms relative to standalone firms. Our outcomes combine ownership structure (100%-owned vs. partially owned group) and strategic configuration (single-segment, multi-segment with vertical integration, or multi-segment with diversification). *RECOVERY* PPE_{t-3} is the ratio of PPE over total assets multiplied by the recovery rate of the accounting value PPE investments in case of liquidation lagged 3 years; ln $ASSETS_{t-3}$ is the natural logarithm of total assets lagged 3 years; *ID(%Owner = 100%)* is a dummy equal to 1 if the largest shareholder owns 100% of the firm. All models include fixed effects at the NACE 2-digit industry level and by country; we clustered standard errors at the country level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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organizational forms facilitate the coordination and protection of investments without requiring full ownership.

In sum, our findings confirm that the organizational form varies with the degree of ownership concentration, as supported by PRT arguments. Full ownership tends to adopt standalone firm to enforce property rights control directly, whereas when decoupling control from ownership, BGs enable more flexible capital arrangements to manage property rights, despite the BG being more costly (i.e., information requirements and formal governance mechanisms for each affiliate).

Robustness Checks

To validate the robustness of our core findings, Table VII presents complementary analyses referring to Hypotheses 1 and 2. The analysis in panels A and B, to test Hypothesis 1, substitutes RECOVERY PPE as a proxy for firm-specific investments used in Table V. In order to ensure that our results are not driven by a single operationalization of asset specificity, we also use two alternative proxies. The first proxy is INTANGIBLES, usually more firm-specific (and less contractable) than tangible assets. The second proxy, SPEED DEPRECIATION, draws on Kermani and Ma (2023), who argue that the speed of asset depreciation can also reflect the redeployability of investments (i.e., their specificity). Concretely, when fixed assets are more firm-specific, they tend to depreciate more quickly due to their lower potential for alternative use. In Panel A in Table VII, we replace *RECOVERY PPE* with *INTANGIBLES* measured as the ratio of intangible assets to total assets. The results strongly support Hypothesis 1. Across all strategic orientations and size categories, the coefficients on *INTANGIBLES* are generally positive and statistically significant. These results mirror the earlier findings with *RECOVERY PPE*, confirming the robustness of the link between assets' specificity and the organizational choice.

In Panel B in Table VII, we replace *RECOVERY PPE* with *SPEED_DEPRECIATION*, measured as the ratio of annual amortization to fixed assets. Again, the results strongly support Hypothesis 1. All the coefficients on Speed of Depreciation are positive and statistically significant. For example, among medium-sized, vertically-integrated firms, the coefficient is 0.182 ($p < 0.01$), while for medium-sized diversified firms it reaches 0.191 ($p < 0.01$). Importantly, the significance and direction of these effects persist across different firm types, confirming the robustness of the link between specificity and organizational choice.

Following Ziliak and McCloskey (2008), we evaluate our results not only through statistical significance but also through economic magnitude. Because our alternative proxies are measured on different scales, we do not interpret coefficient magnitudes in isolation. Instead, for each proxy and subsample we compute an economically scaled effect as the product of the estimated coefficient and the sample mean of the corresponding regressor. Performing this exercise across the different subsamples considered in the paper leads to the same conclusion; that is, the RECOVERY PPE is not only conceptually closer to our notion of easily contractible assets (as it reflects collateral value upon liquidation), but it also delivers a larger economically scaled impact than either INTANGIBLES or SPEED DEPRECIATION.

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The analysis of Hypothesis 2 tests the effects of controlling ownership on the likelihood of adopting a BG (Hypothesis 2), with the dummy variable *ID(%Owner = 100%)*, that captures the full control by the owner. While this measure simplifies interpretation, it may obscure variation in control intensity across firms with high (but not full) ownership. To address this, Panel C in Table VII estimates the same multinomial logit model using *%OWNERSHIP*, a continuous variable capturing the precise equity investment held by the main owner (in standalone firms) or in the parent company (in BGs).

Our results continue to support Hypothesis 2. The coefficients for *%OWNERSHIP* are consistently negative and significant, especially for the partially owned BG category. This implies that, as ownership concentration increases, firms are more likely to remain standalone rather than adopt a BG, particularly one involving partial ownership. For instance, in medium-sized diversified firms, the coefficient for the partially owned group is −7.10 ($p < 0.01$), while in large-sized vertically integrated firms, it is −8.25 ($p < 0.01$). For fully owned BGs, the pattern is also negative and significant, though with slightly smaller magnitudes, consistent with the theoretical prediction that fully owned BGs require more investment to maintain control than partially owned ones.

The use of a continuous control variable thus reinforces our conceptual framework, showing that the probability of adopting a particular organizational form reflects a nuanced balancing of control and flexibility when exercising the property rights. Together, Panels A, B, and C in Table VII confirm that our core results are not sensitive to variable operationalization.

As an additional robustness check, we have re-estimated our baseline specifications using alternative lag structures for the explanatory variables. Specifically, we lag all explanatory variables by 1 and 2 years ($t-1$ and $t-2$) instead of $t-3$. The results remain qualitatively unchanged: coefficient signs, magnitudes, and statistical significance are consistent with those reported in the main analyses, both for the estimation of Equation (1) and for the specification in Equation (2).

DISCUSSION AND CONCLUSION

This study's central contribution is a theory of property-rights architecture. Building on the design decision PRT already answers, who should own a given asset (Grossman and Hart, 1986; Hart and Moore, 1990), we show that a second decision, whether the owned assets are held within one legal entity or partitioned across several, is itself governed by property-rights logic, that is, the choice between the standalone firm and the BG. Our theory goes further than TCE. In particular, we hold the organizational boundary decision (i.e., focused, vertical-integrated, or diversified strategies), and therefore transaction-risk constant, and focus on the controlling owners' decision about the organizational form: the BG organizational form over a standalone form when firm-specific assets are prevalent, or ownership is, or might be, shared. This is an argument that TCE's transaction-hazard logic does not, on its own terms, predict. Our approach predicts, instead, that even holding transaction risks constant, firms

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Table VII. Robustness tests for Hypotheses 1 and 2

	Single segment: Focused			*Multiple segments: Vertical integration*			*Multiple segments: Diversification*		
	Small	*Medium*	*Large*	*Small*	*Medium*	*Large*	*Small*	*Medium*	*Large*
Panel A: Hypothesis 1 with INTANGIBLES as a measure of Asset Specificity Reference Group: STANDALONE									
$INTANGIBLES_{t-3}$	0.018	0.172***	0.432***	0.306***	0.322***	0.275***	−0.015	0.696***	0.286 (0.228)
	(0.023)	(0.048)	(0.123)	(0.111)	(0.090)	(0.087)	(0.107)	(0.119)	
$lnASSETS_{t-3}$	0.011***	0.022***	0.042***	0.027***	0.033***	0.039***	0.028**	0.039**	0.073***
	(0.002)	(0.004)	(0.006)	(0.007)	(0.004)	(0.010)	(0.011)	(0.015)	(0.020)
N. Observations	431,400	108,010	16,233	213,966	100,231	25,189	92,271	31,700	5633
R^2	0.026	0.059	0.148	0.664	0.638	0.512	0.374	0.382	0.415
Panel B: Hypothesis 1 with SPEED DEPRECIATION as a measure of asset specificity									
$SPEED\ DEPRECIATION_{t-3}$	0.011***	0.040***	0.038**	0.088***	0.182***	0.160***	0.094***	0.191***	0.223***
	(0.002)	(0.007)	(0.016)	(0.012)	(0.015)	(0.031)	(0.011)	(0.020)	(0.041)
$lnASSETS_{t-3}$	0.013***	0.025***	0.048***	0.036***	0.046***	0.088***	0.052***	0.070***	0.097***
	(0.001)	(0.002)	(0.004)	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)	(0.007)
N. Observations	411,269	106,544	16,556	108,332	45,373	9192	80,786	28,588	5299
R^2	0.022	0.039	0.086	0.229	0.268	0.534	0.211	0.232	0.315
Panel C: Hypothesis 2 with % ownership. Reference group: STANDALONE									
100%-OWNED BG									
$RECOVERY\ PPE_{t-3}$	−5.083***	−4.183***	−2.125**	−5.317***	−5.315***	−4.874***	−3.724***	−3.511***	−1.460 (1.237)
	(0.528)	(0.663)	(0.867)	(1.447)	(1.574)	(1.001)	(0.947)	(1.298)	
$lnASSETS_{t-3}$	0.789***	0.550***	0.380***	1.368***	1.436***	1.312***	1.264***	1.138***	0.877***
	(0.074)	(0.056)	(0.061)	(0.083)	(0.073)	(0.063)	(0.095)	(0.076)	(0.049)
% OWNERSHIP	−2.590***	−2.212***	−2.032***	−3.602***	−3.799***	−4.592***	−3.846***	−3.794***	−5.201***
	(0.260)	(0.377)	(0.715)	(0.543)	(0.428)	(0.359)	(0.475)	(0.455)	(0.583)

(Continues)

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Table VII. (Continued)

	Single segment: Focused			*Multiple segments: Vertical integration*			*Multiple segments: Diversification*		
	Small	*Medium*	*Large*	*Small*	*Medium*	*Large*	*Small*	*Medium*	*Large*
PARTIALLY OWNED BG									
RECOVERY PPE_{t-3}	−4.269*** (0.362)	−3.620*** (0.644)	−2.221*** (0.632)	−2.787* (1.461)	−2.437 (1.679)	−3.000** (1.448)	−2.662*** (0.745)	−3.082*** (0.921)	−0.659 (1.327)
$lnASSETS_{t-3}$	0.731*** (0.090)	0.380*** (0.073)	0.263*** (0.072)	1.161*** (0.103)	1.144*** (0.100)	0.999*** (0.062)	1.217*** (0.117)	0.954*** (0.118)	0.718*** (0.067)
% OWNERSHIP	−4.954*** (0.315)	−4.440*** (0.442)	−3.770*** (0.873)	−6.139*** (0.659)	−6.350*** (0.462)	−8.251*** (0.718)	−6.956*** (0.695)	−7.105*** (0.661)	−10.921*** (1.361)
N. Observations	227,390	50,908	7137	74,758	33,407	6576	56,623	20,197	3383

Note: This table presents robustness checks using alternative measures of asset specificity and control preferences. Panel A tests Hypothesis 1 using *INTANGIBLES* (ratio of intangibles to total assets) as an alternative proxy for asset specificity. Panel B tests Hypothesis 1 using *SPEED DEPRECIATION* (ratio of depreciation to total fixed assets) as an alternative proxy for asset specificity. All models in Panels A and B include fixed effects at the NACE 2-digit level and country-fixed effects. We cluster standard errors at the country level. Panel C tests Hypothesis 2 using *% OWNERSHIP* (percentage of equity held by the largest shareholder) instead of the binary ownership concentration indicator. All models in Panel C include NACE 2-digit fixed effects only, and standard errors are clustered at the country level. We estimate all regressions in both panels separately by strategic configuration (single-segment, multi-segment with vertical integration, and multi-segment with diversification) and by firm size (small: 10–49 employees; medium: 50–249; large: 250+). ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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adopt the BG form over an integrated standalone form when the proportion of firm-specific assets is high or when ownership is, or might be, shared. Partitioning control across different affiliates, rather than concentrating it in one entity, is precisely what facilitates owners the exercise and protection of the use, appropriate, and transfer rights over the assets. In this sense, our contribution to PRT is to incorporate the organizational form itself as an endogenous outcome of the theory, and not only the allocation of rights over individual assets. Legal partitioning is therefore not merely a legal shell surrounding allocation of rights; as our discussion of firm-specific assets and ownership sharing above shows, it is itself a mechanism for structuring the exercise of use, appropriation, and transfer rights.

BGs are ubiquitous around the world and capture a relevant portion of economic activity everywhere (Aguilera et al., 2023). Institutional voids theory explains an important part of this pattern, showing how BGs fill gaps in weak market institutions through vertical integration (Dau et al., 2021; Khanna and Yafeh, 2005, 2007; Yiu et al., 2005). This explanation, however, leaves open why standalone firms also vertically integrate (Fan et al., 2017), why vertically integrated BGs persist in developed markets (Atalay et al., 2014), and why BGs also pursue diversification strategies (e.g., Guillén, 2000; Khanna and Yafeh, 2007). Alternative explanations of BGs based on political influence and tunnelling strategies do not explain why BGs remain common even among small firms with limited political influence, also in developed economies (Iacobucci, 2002; Morck, 2010). The PRT serves as a useful conceptual framework to analyse BG prevalence in developing and developed economies regardless of the different strategies used.

Our PRT Arguments

According to Foss and Foss (2022), firms combine easily contractible assets with those that require specific, non-contractible investments, with ownership serving as the key mechanism governing their joint use. Our extension of the PRT shows that a higher proportion of firm-specific assets constrains the exercise and protection of controlling owners' property rights through standalone organizational forms, increasing the salience of BG structures, particularly under shared ownership (H1). Conversely, entities that are fully owned by controlling owners and characterized by a low proportion of firm-specific assets are expected to be organized as standalone firms (H2).

PRT emphasizes that control over assets determines how effectively owners can use, appropriate, and transfer surplus generated by firm-specific assets. Firm-specific assets are central to competitive advantage, and controlling owners have incentives to adopt organizational forms that better protect their property rights. First, BGs allow allocation of firm-specific assets across legally distinct entities, enabling modular and flexible deployment while insulating these assets from external claims. Second, BGs enable owners to determine which affiliates hold claims over specific assets and to redistribute surplus internally through capital markets and transfer pricing. Third, BGs facilitate the reallocation of non-contractible assets by reallocating ownership across multiple legal entities, allowing the joint transfer of complex assets and complementary production factors that cannot be efficiently governed through contracts.

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Ownership sharing further reinforces these property-rights arguments. When owners share or may share ownership, trade-offs between control, risk exposure, and financing become salient. BGs, through pyramidal ownership and partial ownership arrangements, allow controlling owners to retain effective control while diversifying risk and limiting liability. Compared with standalone firms, BGs offer greater flexibility and protection in exercising use, appropriation, and transfer rights under shared ownership, including the ability to reallocate assets, manage risk across affiliates, and transfer ownership without relinquishing control.

Overall, despite higher governance and information costs, the BG organizational form offers controlling owners organizational modularity to exercise and protect property rights over firm-specific assets, particularly when these assets are central to value creation and when ownership is shared.

Empirical Design and Findings

Our sample consists of 30,355 BGs and 654,192 standalone firms. We analyse the determinants of the adoption of BG versus standalone based on asset specificity and ownership sharing, splitting samples by their strategy configurations: focused on a single-segment, multi-segment diversification, or multi-segment in a vertically integrated structure. We further split our sample according to size to capture the differences in decisions on organizational form conditioned by complexity. Additionally, to be consistent with previous findings, we control for institutional context by analysing only BGs in which all of their affiliates are located within the same country, and by splitting the sample into high-income and low-income countries. Our results are highly consistent with our predictions. Firms with higher asset specificity are significantly more likely to adopt a BG organizational form, supporting Hypothesis 1. This effect tends to be stronger for larger firms, especially in high-income countries with well-developed institutions. In developing countries, where weak market institutions make market transactions difficult, it is not that relevant whether assets are firm-specific (i.e., non-easily contractible), since all assets suffer from contractability restrictions. Related to Hypothesis 2, firms in which the controlling owner holds 100 per cent ownership are significantly less likely to adopt a BG. This finding suggests that concentrated ownership is associated with standalone organizational forms for safeguarding property rights, whereas shared ownership is associated with BGs, especially those with partially owned affiliates, that facilitate flexible coordination and property-rights protection.

Our study makes three key contributions to the strategic management and international business literature. First, we advance PRT theoretically: we show that organizational form itself, BG versus standalone firm, is an endogenous property-rights design decision, analytically distinct from, though possibly jointly determined with, who should own a given asset, the question the new PRT already addresses. Differences in asset specificity and ownership sharing shape when this partitioning occurs, with BG structures providing a more effective mechanism despite higher governance and information costs in some configurations of these factors. Second, we offer empirical evidence from a large global dataset that spans developed and emerging economies

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and includes both private and publicly listed firms. Using an ownership-based identification methodology of BGs, we trace full chains of control to distinguish between fully and partially owned affiliates, enabling a more fine-grained analysis of ownership structures. Third, we contribute to the strategic management literature by broadening the type of strategies examined through a PRT lens. Moving beyond the dominant focus on vertical integration, we show that our findings hold across the three types of strategies (focused, vertically integrated, and diversified), thereby offering a more nuanced understanding of BGs.

Future Research and Other Applications of our PRT Arguments

Our research is subject to several limitations that also point to avenues for future research. First, we deliberately restrict our analysis to domestic BGs (i.e., those without foreign affiliates), to avoid potential distortions arising from the creation of foreign affiliates for legal or regulatory purposes and to isolate institutional context effects more clearly. While this research design enhances internal validity, it excludes multinational BGs from our sample. Because internationalization processes may shape the articulation of economic activity through BGs, particularly where host-country regulations require legally independent affiliates (Aguilera et al., 2020; Du et al., 2011), future research could analyse how property rights influence expansion through foreign affiliates.

Second, our analysis relies on proxies for asset specificity and owners sharing. We measure asset specificity using the recovery rate of PPE investments, based on the average liquidation values from Kermani and Ma (2023), which might vary across institutional contexts. We validate our findings using alternative measures (asset intangibility and the speed of depreciation), yielding consistent results. Nonetheless, further research could develop more precise, context-specific measures of liquidation values and asset specificity.

Third, measuring ownership and control sharing is inherently more complex in BGs. We distinguish between BGs with fully owned affiliates and those with partial ownership. While control sharing is straightforward to identify in fully owned BGs, indirect control across tiers in partially owned BGs complicates measurement. Further work is needed to develop more refined measures of ownership and control sharing along complex BGs.

Fourth, although we mitigate causality concerns using lagged explanatory variables, data limitations restrict the use of more robust methods to fully address endogeneity. Future studies could apply techniques, such as necessary condition analysis (Dul, 2016), for example, to assess whether specific institutional conditions are required for BGs and standalone firms to differ in the management and protection of property rights. In addition, while our theoretical and empirical approach conditions organizational form (BG vs. standalone firm) on deployed strategies (e.g., focused, vertical integration, and diversification), we do not examine their joint determination. Further research could disentangle the directionality between organizational forms and types of strategies.

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Fifth, our design contrasts two specific ownership arrangements. Future research could extend our property-rights logic to understand the advantages of ownership modularity in more general situations: whenever different agents decide to collaborate. For example, once two firms decide to collaborate, our theoretical expansion of the PRT could help in understanding when ownership modularity (e.g., a joint venture formalized as a legally independent entity, generally implying firm-specific assets and ownership sharing) offers a better solution than alternatives without ownership modularity (e.g., a collaboration contract where all investments are provided by one of the corporations). The same logic applies to other arrangements built on a legal-ownership partitioning decision, such as special purpose vehicles, and carve-outs, each an instance of the same underlying choice between concentrating and partitioning property rights across different legal entities.

Finally, our analysis focuses on firm-specific assets that are localized within individual affiliates or organizational entities (i.e., divisions). However, many firm-specific assets such as organizational knowledge or owners' competence, are readily transferable across tasks or units within an organization (Atalay et al., 2014; Foss et al., 2021; Teece, 1980). PRT also favours the BG organizational form when such specific but transferable assets within the organization are critical. Further research should examine the relevance of these assets for the organizational decision (BGs or standalone firm) under the PRT lenses, and to develop proxies that more accurately capture their relevance.

In conclusion, our study does not simply apply PRT to explain the choice between organizing economic activity as a BG or a standalone firm; rather, we develop a theory of property-rights architecture, showing that legal-entity partitioning is itself a second, endogenous property-rights design decision alongside ownership, and giving the new PRT the conceptual room to explain this organizational choice. By focusing on asset specificity and ownership shared control, and accounting for firm strategies and institutional contexts, we offer a novel conceptual framework to explain the organizational logic behind the formation of BGs, one that complements and extends the traditional explanation rooted in institutional voids to other institutional settings. Our large and diverse dataset enables us to reveal that BGs are not only a response to institutional inefficiencies, but also an effective governance structure to manage property rights when firm-specific assets are present, especially in shared-ownership structures. Our findings underscore that BGs are not an anomaly of emerging markets, but a robust and rational organizational form observed across income levels, firm size, and strategic types. More broadly, as property-rights architecture is a genuine construct of PRT rather than a feature specific to BGs, its logic should extend beyond BGs to other arrangements built on the same legal-ownership partitioning decision (e.g., joint ventures). We hope our research opens new avenues for examining BGs as a distinctive and widespread organizational form and encourages future work to further explore their relevance in the theory of the firm, strategic management, and international business.

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NOTES

[1] In our conceptualization, any economic activity may be classified either as a BG or as a standalone firm. For example, conglomerates may be organized as BGs, with different affiliates developing different activities, or as a standalone firm if all the activities are developed by the same legal entity (i.e., multidivisional). Well-known business structures, such as holding companies, may also be classified within this conceptualization, as the apex firm of a BG, with the peculiarity that no other economic activity other than holding shares is developed. Consequently, 'standalone firm' is retained throughout the entire manuscript as the umbrella term for any single legal entity, including internally diversified or vertically integrated conglomerates, while 'business group' denotes any set of legally independent entities under common ownership, regardless of their diversification strategy.

[2] Atalay et al. (2014) show that vertically integrated firms may transact little internally and construct a measure of vertical integration based on these internal transactions. Our cross-country data do not include within-firm/within-group shipment or transaction flows, so we cannot construct a flow-based measure; the affiliate-count measure therefore provides a transparent and feasible proxy that can be applied uniformly across countries.

[3] Kermani and Ma (2023) provide the average PPE accounting value recovery rate in case of liquidation per industry, according to 2-digit SIC code. We use industry conversion tables to transform into the equivalent 2-digit NACE codes. We drop from the sample firms and business groups for which we cannot compute RECOVERY_PPE because the corresponding PPE recovery rate is not available in Kermani and Ma (2023) (e.g., it is not reported for industries with too few observations).

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